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TECHNIQUES FOR PROMOTING PRIVACY WHEN PROVIDING QUERY RESULTS

Abstract

One embodiment sets forth a method for promoting privacy when providing query results. The method can include receiving a query that includes a unique identifier associated with a software application, a date, and at least one parameter, and generating results of the query that include multiple records. The method can further include obtaining a particular random seed value using information included in the query, adjusting the multiple records using noise values generated using multiple hash values based on the particular random seed, and including information based on the adjusted multiple records in a user interface that is output on a display device.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application claims the benefit of U.S. Provisional Application No. 63/561,251, entitled “TECHNIQUES FOR PROMOTING PRIVACY WHEN PROVIDING QUERY RESULTS,” filed Mar. 4, 2024, the content of which is incorporated by reference herein in its entirety for all purposes.

FIELD

[0002] The described embodiments relate generally to providing query results in response to queries. More particularly, the described embodiments provide techniques for promoting privacy when providing the query results.

BACKGROUND

[0003] Privacy protection stands as a significant challenge in the modern-day landscape of data-driven activities. In particular, as data collection and analysis methods become more advanced, there is a growing importance to establish a reasonable balance between the benefits of data utilization and the protection of individuals' privacy rights. One effective strategy in achieving this balance is the addition of noise to data, which is a practice that has been employed across various sectors to, at least to some degree, safeguard sensitive information while maintaining data utility.

[0004] The necessity to add noise to data can arise from the need to anonymize or obfuscate sensitive information contained within datasets. For example, healthcare records, financial transactions, and social science surveys often include personal details such as medical histories, financial transactions, and demographic information, respectively, such that it can be desirable to implement privacy measures. By injecting controlled noise into datasets, the original data's statistical properties can remain largely intact—with individual-level details becoming largely obscured—thereby making it significantly more challenging for malicious parties to identify specific individuals (while preserving the overall usefulness of the datasets).

[0005] Importantly, while noise addition techniques can offer some level of privacy protection, they are not immune to potential vulnerabilities. In particular, current implementations of noise addition, particularly those using simple randomization or fixed perturbation methods, can be susceptible to reverse engineering attacks. In this regard, sophisticated adversaries can potentially exploit patterns in the noise, combine multiple datasets to infer sensitive information, and so on, to effectively glean information that can compromise individuals' privacy despite the added noise. For example, generating random noise for repeated queries/query results can, in some cases, enable a malicious entity to average the noise and thereby circumvent the intended protections.

[0006] Accordingly, what is needed are improved techniques for providing relevant query results to users in response to their queries.

SUMMARY

[0007] The aspects described herein relate generally to providing query results in response to queries. More particularly, the described embodiments provide techniques for promoting privacy when providing the query results.

[0008] One aspect sets forth a method for promoting privacy when providing query results. According to some embodiments, the method can be implemented by a computing device, and includes the steps of receiving at least one query that includes a unique identifier associated with a software application, a date, and at least one parameter, generating query results based on the query, wherein the query results include one or more records, obtaining a particular random seed value based on the unique identifier and the date, generating a plurality of hash value using the unique identifier, the particular random seed value, and at least one aspect of a corresponding

record of the one or more records, generating a plurality of noise values for corresponding hash values of the plurality of hash values, adjusting the one or more records using the plurality of noise values to produce one or more adjusted records, and causing information based at least in part on the one or more adjusted records to be included in a user interface that is output on a display device.

[0009] According to some aspects, obtaining the particular random seed value based on the unique identifier and the date includes: identifying a respective plurality of random seed values that corresponds to the date, assigning the unique identifier of the software application to a dividend value, assigning a number of random seed values in the respective collection of random seed values to a divisor value, and performing a modulo operation on the dividend value and divisor value, where a remainder of the modulo operation corresponds to a position of the particular random seed value relative to other ones of the respective plurality of random seed values.

[0010] According to some aspects, for a given record of the one or more records, the at least one aspect includes: a metric name that corresponds to the record, a date value that corresponds to the record, a period value that corresponds to the record, a grouping set value that corresponds to the record, or some combination thereof.

[0011] According to some aspects, generating a given respective noise value for a given respective hash value comprises: generating a random value based on the given respective hash value, and (ii) a personalization value that is based at least in part on the unique ID, and (2) utilizing the random value to obtain the given respective noise value.

[0012] According to some aspects, the given respective hash value is a uniformly distributed random value, the random value is a normalized random value, and the given respective noise value is a Gaussian distribution value (e.g., discrete, continuous, etc.).

[0013] According to other aspects, the given respective hash value is generated using a SHA-256 hash function, the random value is generated using a SHA-256 hash function (or alternative hash function), and the Gaussian distribution value is based on a mean of zero and a standard deviation of two.

[0014] According to some aspects, adjusting a given record of the one or more records based at least in part on the respective noise value to produce the corresponding adjusted record comprises: performing at least one operation on the given record based at least in part on the respective noise value and a type of the given record.

[0015] Other aspects include a non-transitory computer readable storage medium configured to store instructions that, when executed by a processor included in a computing device, cause the computing device to carry out the various steps of any of the foregoing methods. Further aspects include a computing device that is configured to carry out the various steps of any of the foregoing methods.

[0016] Other advantages of the aspects described herein will become apparent from the following detailed description taken in conjunction with the accompanying drawings which illustrate, by way of example, the principles of the described embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] The included drawings are for illustrative purposes and serve only to provide examples of possible structures and arrangements for the disclosed inventive apparatuses and methods for providing wireless computing devices. These drawings in no way limit any changes in form and detail that may be made to the embodiments by one skilled in the art without departing from the spirit and scope of the embodiments. The embodiments will be readily understood by the following detailed description in conjunction with the accompanying drawings, wherein like reference

numerals designate like structural elements.

[0018] FIG. 1 illustrates a block diagram of different components of a system that can be configured to implement the various techniques described herein, according to some embodiments.

[0019] FIG. 2 illustrates a conceptual diagram that illustrates distinctions between noised and non-noised records displayed within table-based user interfaces, according to some embodiments

[0020] FIG. 3 illustrates a conceptual diagram that illustrates distinctions between noised and non-noised records displayed within a dashboard-based user interface, according to some embodiments

[0021] FIG. 4 illustrates a method for promoting privacy when providing query results, according to some embodiments

[0022] FIG. 5 illustrates a detailed view of a computing device that can be used to implement the various components described herein, according to some embodiments.

DETAILED DESCRIPTION

[0023] Representative applications of apparatuses and methods according to the presently described embodiments are provided in this section. These examples are being provided solely to add context and aid in the understanding of the described embodiments. It will thus be apparent to one skilled in the art that the presently described embodiments can be practiced without some or all of these specific details. In other instances, well known process steps have not been described in detail in order to avoid unnecessarily obscuring the presently described embodiments. Other applications are possible, such that the following examples should not be taken as limiting.

[0024] The described embodiments relate generally to providing query results in response to queries. More particularly, the described embodiments provide techniques for promoting privacy when providing the query results.

[0025] FIG. 1 illustrates a block diagram of different components of a system **100** that can be configured to implement the various techniques described herein, according to some embodiments. As shown in FIG. 1, the system **100** can include a client computing device **102** and a server computing device **112**. It is noted that, in the interest of simplifying this disclosure, the client computing device **102** and the server computing device **112** are typically discussed in singular capacities. In that regard, it should be appreciated that the system **100** can include any number of client computing devices **102** and server computing devices **108**, without departing from the scope of this disclosure.

[0026] According to some embodiments, the client computing device **102** and the server computing device **112** can represent any form of computing device operated by an individual, an entity, etc., such as a wearable computing device, a smartphone computing device, a tablet computing device, a laptop computing device, a desktop computing device, a gaming computing device, a smart home computing device, an Internet of Things (IoT) computing device, a rack mount computing device, and so on. It is noted that the foregoing examples are not meant to be limiting, and that the client computing device **102**/server computing device **112** can represent any type, form, etc., of computing device, without departing from the scope of this disclosure.

[0027] Under an example scenario, the server computing device **112** can implement (or can communicate with other computing devices that implement) an online software application store (also referred to herein as an “App Store”) that manages software applications **118** and the distribution thereof onto computing devices. In this regard, the server computing device **112** can be configured to maintain information pertaining to the utilization, distribution, advertisement, etc., of the software applications **118**. In this vein, the client computing device **102** can represent a software development, management, etc., entity that is associated with software applications **118**, and that is seeking to access information associated with the software applications **118** that is managed by, accessible to, etc., the server computing device **112**. According to some embodiments, the client computing device **102** can be associated with (i.e., logged into) a user account (not illustrated in FIG. 1) that is known to the server computing device **112**, where the user account enables the client computing device **102** to access, interact with, etc., the aforementioned

information.

[0028] As shown in FIG. 1, the server computing device **112** can implement a query manager **114** that is configured to receive, process, and respond to queries **104** issued by the client computing device **102** (e.g., via the Internet). According to some embodiments, and as shown in FIG. 1, a given query **104** can include one or more parameters **106** that are interpreted by the query manager **114** to provide query results (also referred to herein as “records”). The parameters **106** can include any amount, type, form, etc., of information, at any level of granularity, consistent with the scope of this disclosure.

[0029] As shown in FIG. 1, the server computing device **112** can manage, access, etc., a database **116** (e.g., stored on one or more local storage devices, one or more network storage devices, one or more cloud-based storages, etc.). According to some embodiments, the database **116** can manage data **122** for software applications **118**, where each software application **118** is associated with a unique identifier **120**. The data **122** for a given software application **118** can include any information related to the advertisement, distribution, utilization, etc., of the software application **118**. It is noted that the foregoing examples are not meant to be limiting, and that the data **122** for a given software application **118** can include any amount, type, form, etc., of information, at any level of granularity, consistent with the scope of this disclosure.

[0030] Accordingly, the query manager **114** can receive queries **104** from the client computing device **102**, interface with the database **116** to obtain information based on the parameters **106**, process the information (where necessary), and return query results to the client computing device **102**. According to some embodiments, the query results can be organized, formatted, etc., in a manner that is understood by the client computing device **102**. In turn, the client computing device **102** can display the query results using the appropriate applications, user interfaces, etc., to enable a user of the client computing device **102** to interact with the query results.

[0031] As described herein, malicious entities can, in some cases, effectively glean private information about sensitive aspects of the software application **118** (even when the aforementioned query results do not directly include such sensitive information). Such sensitive information can include, for example, user-specific information, device-specific information, and so on. In this regard, it can be beneficial to add noise to the aforementioned query results (illustrated in FIG. 1 as noised query results **128**) in the interest of promoting privacy.

[0032] In order to effectively provide noised query results **128**, the query manager **114** can manage one or more collections of random seed values **124**, where a given collection of random seed values includes a plurality of random seed values. According to some embodiments, each collection of random seed values **124** can correspond to a period of time (e.g., a week, a month, a year). In this vein, a new collection of random seed values **124** can be generated when appropriate, e.g., at the start of a new period of time. For example, if a collection of random seed values **124** is generated on a monthly basis, then a new collection of random seed values **124** can be generated and added on the first day of each month (where previous collections of random seed values **124** are preserved). It is noted that the foregoing examples are not meant to be limiting, and that any number of collections of random seed values **124** can be managed, and that the collections of random seed values **124** can include any number of random seed values **124**, and can be generated on any periodic, conditional, etc., basis, at any level of granularity, consistent with the scope of this disclosure.

[0033] According to some embodiments, each random seed value **124** (in a given collection of random seed values **124**) can represent a large entropy (e.g., a two-hundred fifty-six (256) bit) number, which can be generated using a cryptographically secure hash function (e.g., SHA-256). It is noted that the foregoing examples are not meant to be limiting, and that the random seed value **124** can represent any type, size, form, etc., of random seed value, and can be generated using any function, consistent with the scope of this disclosure. According to some embodiments, each random seed value **124** (in a given collection of random seed values **124**) can be assigned an

identifier (not illustrated in FIG. 1) that represents the position of random seed value **124** relative to other random seed values **124** in the given collection of random seed values **124**. For example, a first random seed value of random seed values **124** can take on the identifier “1”, a second random seed value of random seed values **124** can take on the identifier “2”, and so on. It is noted that the foregoing examples are not meant to be limiting, and that the random seed values **124** can be assigned any type, form, etc., of identifier, at any level of granularity, consistent with the scope of this disclosure.

[0034] According to some embodiments, when a query **104** is received, the parameters **106** can include a unique identifier **120** (of a software application **118** in which a user is interested in obtaining information therefor), a date (e.g., a relevant date, date range, etc., that corresponds to the information being sought), and any other parameters that inform the query manager **114** about the records that should be obtained, generated, and so on. The query manager **114** can then interface with the database **116** to establish one or more (non-noised) records.

[0035] When the records have been obtained, the query manager **114** can implement techniques that add noise to the records. In particular, the query manager **114** can identify the collection of random seed values **124** that corresponds to the date provided in the query **104**. For example, if the date specifies Jan. 1, 2024, then the query manager **114** can identify the collection of random seed values **124** that corresponds to the month of January in 2024 (e.g., when collections of random seed values **124** are generated on a monthly basis). In another example, if the date specifies a range (e.g., Feb. 1, 2024 through Feb. 15, 2024), then the query manager **114** can identify the collection of random seed values **124** that corresponds to the starting date of the range (i.e., Feb. 1, 2024). It is noted that the foregoing examples are not meant to be limiting, and that collections of random seed values **124** can be managed, identified, etc., in accordance with any amount, type, form, etc., of information, at any level of granularity, consistent with the scope of this disclosure.

[0036] Next, the query manager **114** identifies the random seed value **124** (within the identified collection of random seed values **124**) that corresponds to the unique identifier **120**. To identify the random seed value **124**, the query manager **114** (1) assigns the unique identifier **120** of the software application—e.g., “23382907532”—to a dividend value, and (2) assigns a number of random seed values **124** in the collection of random seed values **124**—e.g., “1000”—to a divisor value. In turn, the query manager **114** performs a modulo operation on the dividend value (“23382907532”) and the divisor value (“1000”), to obtain a remainder value (i.e., $23382907532 \% 1000 = 532$), where the remainder value (“532”) corresponds to the position of the random seed value **124** relative to other random seed values **124** (in the collection of random seed values **124**). It is noted that the foregoing examples are not meant to be limiting, and that queries **104** can be mapped to random seed values **124** using any, type, form, etc., of approach, at any level of granularity, consistent with the scope of this disclosure. In any case, the random seed value **124** can then be obtained from the collection of random seed values **124**.

[0037] When the random seed value **124** is obtained, the query manager **114** can generate, for each record of the one or more records (initially gathered in accordance with the query **104**), a respective hash value based on (1) the unique identifier **120** (included in the query **104**), (2) the random seed value **124**, and (3) at least one aspect of the record. According to some embodiments, the at least one aspect can include a metric name that corresponds to the record (e.g., advertisement, distribution, utilization, etc., metrics associated with the software applications **118**), a date value that corresponds to the record (e.g., a starting date, a date range, etc.), a period value that corresponds to the record (e.g., a reporting period), a grouping set value that corresponds to the record (e.g., specification(s) of grouped columns used to aggregate data), and the like. It is noted that the foregoing examples are not meant to be limiting, and that the aforementioned aspects can include any amount, type, form, etc., of information associated with, relevant to, etc., a given record, at any level of granularity, consistent with the scope of this disclosure.

[0038] According to some embodiments, to generate a respective hash value for a given record, the

aforementioned (1) unique identifier **120**, (2) random seed value **124**, and (3) at least one aspect of the record can be provided to a cryptographically secure hash function (e.g., SHA-256), which then outputs the respective hash value. Again, other hash functions can be utilized to generate hash values for records without departing from the scope of this disclosure. For example, the hash function can be selected (e.g., from a collection of available/distinct hash functions) based on the aforementioned (1) unique identifier **120**, (2) random seed value **124**, and (3) at least one aspect of the record, such that one or more hash functions can be utilized to generate the respective hash values for a set of records. It is noted that the foregoing examples are not meant to be limiting, and that any number, type, form, etc., of hash function(s) can be utilized to generate hash values for records, consistent with the scope of this disclosure.

[0039] According to some embodiments, the hash values output by the aforementioned hash function are uniformly-distributed random values, meaning they are equally capable of taking on a value of zero to the maximum value that can be output by the utilized hash function. Notably, given a hash value (i.e., a hash function output) is produced using a random initial seed value (and, in some cases, other values), the hash value is also a random number that cannot be predicted or recovered without knowing the random initial seed value (and other values, when utilized).

[0040] It can be beneficial to effectively establish normalized random values (i.e., values following a uniform distribution) to be used to perform noising adjustments to the records that are based at least in part on the hash values. To implement this approach, the query manager **114** can, for a given hash value, generate a normalized random value based on (i) the given hash value—and, optionally, (ii) a personalization value that is based at least in part on the unique ID (and/or any other information). The normalized random value can be generated, for example, using one or more random number generators that receive the hash value (and other optional parameters) as input(s), and then output the normalized random value. According to some embodiments, the one or more random number generators can be deterministic, such that the same input value(s) always result in the same output value. It is noted that the foregoing examples are not meant to be limiting, and that a given normalized random value can be based on any amount, type, form, etc., of information, at any level of granularity, and can be based on any number, type, form, etc., of functions, consistent with the scope of this disclosure.

[0041] When the query manager **114** has generated respective normalized random values for the records, the query manager **114** generates respective noise values to be used to adjust the corresponding records. According to some embodiments, the query manager **114** utilizes a given normalized random value as a basis for obtaining a corresponding Gaussian distribution value (e.g., discrete, continuous, etc.) that conforms with particular parameters (e.g., a mean of zero (0), a standard deviation of two (2), etc.), where the Gaussian distribution value effectively functions as the respective noise value. For example, a corresponding Gaussian distribution value can be obtained by passing a given normalized random value through an inverse cumulative distribution function of the desired noise distribution. In some cases, the Gaussian distribution value can also be adjusted, based on, etc., any aspect of the record (and/or other information) to which the Gaussian distribution value will be utilized to apply noise. The Gaussian distribution value can also be adjusted based on one or more rules **126**, which are described below in greater detail. It is noted that any number of approaches can be utilized to obtain a Gaussian distribution value based on a given normalized random value, without departing from the scope of this disclosure.

[0042] Additionally, and according to some embodiments, the query manager **114** can reference a set of rules **126** that can influence how the respective noise values are applied to the records, how the records are adjusted (e.g., independent from the noise values, or in addition thereto), and so on. For example, one or more rules **126** can promote crowd anonymity interests, e.g., for records having identifying records (e.g., operating system (OS) versions, platform versions, etc.), such that if the software application **118** has not been used on a given day, week, month, etc., by five (5) distinct devices, information for those records will be empty. In another example, one or more rules

126 can apply to daily, weekly, monthly, etc., records that constitute aggregations of sensitive information (e.g., a sessions report that identifies a number of unique devices, a number of sessions, total session duration, etc.). In yet another example, one or more rules **126** can apply to money-based records (individual, aggregated, etc.), given such values are typically constituted from discrete price amounts. It is noted that the foregoing examples are not meant to be limiting, and that the rules **126** can include any amount, type, form, etc., of rules, at any level of granularity, consistent with the scope of this disclosure.

[0043] In any case, when the noise values/records have been effectively obtained, adjusted, etc., in accordance with the techniques described herein, the query manager **114** can adjust each record in accordance with its respective noise value. When the query manager **114** has completed adding noise to the records, the query manager **114** can package the noised records (illustrated in FIG. **1** as noised query results **128**) and provide the noised query results **128** to the client computing device **102** (that issued the query **104**), and/or to any other devices that should receive the noised query results **128**. In turn, the client computing device **102** (and/or other devices) can cause information based at least in part on the noised query results **128** to be displayed in a user interface (UI) that is output on a display device. Examples of such user interfaces are illustrated in FIGS. **2-3** and discussed below in greater detail.

[0044] It is noted that the logical breakdown of the entities illustrated in FIG. **1**—as well as the logical flow of the manner in which such entities communicate—should not be construed as limiting. On the contrary, any of the entities illustrated in FIG. **1** can be separated into additional entities within the system **100**, combined together within the system **100**, or removed from the system **100**, without departing from the scope of this disclosure. It is additionally noted that, in the interest of unifying and simplifying this disclosure, the described embodiments primarily pertain to managing data associated with software applications **118**. However, it should be appreciated that the embodiments disclosed herein can be implemented to receive queries **104**—and to provide noised query results **128**—for any type, amount, form, etc., of information, at any level of granularity, consistent with the scope of this disclosure. For example, the unique identifiers **120** described herein can instead represent report identifiers, data set identifiers, and so on, consistent with the scope of this disclosure.

[0045] Additionally, it should be understood that the various components of the computing devices illustrated in FIG. **1** are presented at a high level in the interest of simplification. For example, although not illustrated in FIG. **1**, it should be appreciated that the various computing devices can include common hardware/software components that enable the above-described software entities to be implemented. For example, each of the computing devices can include one or more processors that, in conjunction with one or more volatile memories (e.g., a dynamic random-access memory (DRAM)) and one or more storage devices (e.g., hard drives, solid-state drives (SSDs), etc.), enable the various software entities described herein to be executed. Moreover, each of the computing devices can include communications components that enable the computing devices to transmit information between one another.

[0046] A more detailed explanation of these hardware components is provided below in conjunction with FIG. **5**. It should additionally be understood that the computing devices can include other entities that enable the implementation of the various techniques described herein, without departing from the scope of this disclosure. It should additionally be understood that the entities described herein can be combined or split into additional entities, without departing from the scope of this disclosure. It should further be understood that the various entities described herein can be implemented using software-based or hardware-based approaches, without departing from the scope of this disclosure.

[0047] Accordingly, FIG. **1** provides an overview of the manner in which the system **100** can implement the various techniques described herein, according to some embodiments. A more detailed breakdown of the manner in which these techniques can be implemented will now be

provided below in conjunction with FIGS. 2-5.

[0048] FIG. 2 illustrates a conceptual diagram **200** that illustrates distinctions between noised and non-noised records displayed within table-based user interfaces, according to some embodiments. In particular, a table **202** includes non-noised records that otherwise would be returned by the query manager **114** (in response to a query **104**) if the noising techniques described herein were not applied. Conversely, a table **206** includes noised records (produced by performing noising operations **204** against at least some of the non-noised records included in the table **202**, in accordance with the techniques described herein). As shown in FIG. 2, the Transaction Count values are adjusted in accordance with the noising techniques, thereby obfuscating the information to some degree, and promoting privacy interests.

[0049] FIG. 3 illustrates a conceptual diagram **300** that illustrates distinctions between noised and non-noised records displayed within a dashboard-based user interface, according to some embodiments. In particular, and as shown in FIG. 3, original data **302** is shown, which represents non-noised records that otherwise would be returned by the query manager **114** (in response to a query **104**) if the noising techniques described herein were not applied. Conversely, the noised data **304** is shown, which represents noised records (produced by performing noising operations against the original data **302**, in accordance with the techniques described herein). As shown in FIG. 3, at least some of the Software Application Download count values are adjusted in accordance with the noising techniques, thereby obfuscating the information to some degree, and promoting privacy interests.

[0050] FIG. 4 illustrates a method **400** for promoting privacy when providing query results, according to some embodiments. As shown in FIG. 4, the method **400** begins at step **402**, where the server computing device **112** receives at least one query that includes a unique identifier associated with a software application, a date, and at least one parameter (e.g., as described above in conjunction with FIG. 1).

[0051] At step **404**, the server computing device **112** generates query results based on the query, where the query results include one or more records (e.g., as described above in conjunction with FIG. 1). At step **406**, the server computing device **112** obtains a particular random seed value based on the unique id and the date (e.g., as described above in conjunction with FIG. 1).

[0052] At step **408**, the server computing device **112** generates, a plurality of hash values using the unique identifier, the particular random seed value, and at least one aspect of a corresponding record of the one or more records (e.g., as described above in conjunction with FIG. 1). At step **410**, the server computing device **112** generates a plurality of noise values for corresponding hash values of the plurality of hash values (e.g., as described above in conjunction with FIG. 1).

[0053] At step **412**, the server computing device **112** the one or more records using the plurality of noise values to produce one or more adjusted records (e.g., as described above in conjunction with FIG. 1). At step **414**, the server computing device **112** causes information based at least in part on the one or more adjusted records to be included in a user interface that is output on a display device (e.g., as described above in conjunction with FIG. 1).

[0054] FIG. 5 illustrates a detailed view of a computing device **500** that can be used to implement the various components described herein, according to some embodiments. In particular, the detailed view illustrates various components that can be included in the client computing device **102**, the server computing device **112**, and so on, described above in conjunction with FIG. 1.

[0055] As shown in FIG. 5, the computing device **500** can include a processor **502** that represents a microprocessor or controller for controlling the overall operation of computing device **500**. The computing device **500** can also include a user input device **508** that allows a user of the computing device **500** to interact with the computing device **500**. For example, the user input device **508** can take a variety of forms, such as a button, keypad, dial, touch screen, audio input interface, visual/image capture input interface, input in the form of sensor data, etc. Furthermore, the computing device **500** can include a display **510** (screen display) that can be controlled by the

processor **502** to display information to the user. A data bus **516** can facilitate data transfer between at least a storage device **540**, the processor **502**, and a controller **513**. The controller **513** can be used to interface with and control different equipment through an equipment control bus **514**. The computing device **500** can also include a network/bus interface **511** that couples to a data link **512**. In the case of a wireless connection, the network/bus interface **511** can include a wireless transceiver.

[0056] The computing device **500** also includes a storage device **540**, which can comprise a single disk or a plurality of disks (e.g., SSDs), and includes a storage management module that manages one or more partitions within the storage device **540**. In some embodiments, storage device **540** can include flash memory, semiconductor (solid state) memory or the like. The computing device **500** can also include a Random-Access Memory (denoted as RAM **520**) and a Read-Only Memory (denoted as ROM **522**). The ROM **522** can store programs, utilities, or processes to be executed in a non-volatile manner. The RAM **520** can provide volatile data storage, and stores instructions related to the operation of the computing devices described herein.

[0057] The various aspects, embodiments, implementations, or features of the described embodiments can be used separately or in any combination. Various aspects of the described embodiments can be implemented by software, hardware or a combination of hardware and software. The described embodiments can also be embodied as computer readable code on a computer readable medium. The computer readable medium is any data storage device that can store data that can be read by a computer system. Examples of the computer readable medium include read-only memory, random-access memory, CD-ROMs, DVDs, magnetic tape, hard disk drives, solid state drives, and optical data storage devices. The computer readable medium can also be distributed over network-coupled computer systems so that the computer readable code is stored and executed in a distributed fashion.

[0058] The foregoing description, for purposes of explanation, used specific nomenclature to provide a thorough understanding of the described embodiments. However, it will be apparent to one skilled in the art that the specific details are not required in order to practice the described embodiments. Thus, the foregoing descriptions of specific embodiments are presented for purposes of illustration and description. They are not intended to be exhaustive or to limit the described embodiments to the precise forms disclosed. It will be apparent to one of ordinary skill in the art that many modifications and variations are possible in view of the above teachings.

[0059] As described herein, one aspect of the present technology is the gathering and use of data available from various sources to improve user experiences. The present disclosure contemplates that in some instances, this gathered data may include personal information data that uniquely identifies or can be used to contact or locate a specific person. Such personal information data can include demographics data, location-based data, telephone numbers, email addresses, home addresses, data or records relating to a user's health or level of fitness (e.g., vital signs measurements, medication information, exercise information), date of birth, smart home activity, or any other identifying or personal information. The present disclosure recognizes that the use of such personal information data, in the present technology, can be used to the benefit of users.

[0060] The present disclosure contemplates that the entities responsible for the collection, analysis, disclosure, transfer, storage, or other use of such personal information data will comply with well-established privacy policies and/or privacy practices. In particular, such entities should implement and consistently use privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining personal information data private and secure. Such policies should be easily accessible by users, and should be updated as the collection and/or use of data changes. Personal information from users should be collected for legitimate and reasonable uses of the entity and not shared or sold outside of those legitimate uses. Further, such collection/sharing should occur after receiving the informed consent of the users. Additionally, such entities should consider taking any needed steps for safeguarding and securing

access to such personal information data and ensuring that others with access to the personal information data adhere to their privacy policies and procedures. Further, such entities can subject themselves to evaluation by third parties to certify their adherence to widely accepted privacy policies and practices. In addition, policies and practices should be adapted for the particular types of personal information data being collected and/or accessed and adapted to applicable laws and standards, including jurisdiction-specific considerations. For instance, in the US, collection of or access to certain health data may be governed by federal and/or state laws, such as the Health Insurance Portability and Accountability Act (HIPAA); whereas health data in other countries may be subject to other regulations and policies and should be handled accordingly. Hence different privacy practices should be maintained for different personal data types in each country.

[0061] Despite the foregoing, the present disclosure also contemplates embodiments in which users selectively block the use of, or access to, personal information data. That is, the present disclosure contemplates that hardware and/or software elements can be provided to prevent or block access to such personal information data. For example, the present technology can be configured to allow users to select to “opt in” or “opt out” of participation in the collection of personal information data during registration for services or anytime thereafter. In another example, users can select to provide only certain types of data that contribute to the techniques described herein. In addition to providing “opt in” and “opt out” options, the present disclosure contemplates providing notifications relating to the access or use of personal information. For instance, a user may be notified that their personal information data may be accessed and then reminded again just before personal information data is accessed.

[0062] Moreover, it is the intent of the present disclosure that personal information data should be managed and handled in a way to minimize risks of unintentional or unauthorized access or use. Risk can be minimized by limiting the collection of data and deleting data once it is no longer needed. In addition, and when applicable, including in certain health related applications, data de-identification can be used to protect a user's privacy. De-identification may be facilitated, when appropriate, by removing specific identifiers (e.g., date of birth, etc.), controlling the amount or specificity of data stored (e.g., collecting location data a city level rather than at an address level), controlling how data is stored (e.g., aggregating data across users), and/or other methods.

[0063] Therefore, although the present disclosure broadly covers use of personal information data to implement one or more various disclosed embodiments, the present disclosure also contemplates that the various embodiments can also be implemented without the need for accessing such personal information data. That is, the various embodiments of the present technology are not rendered inoperable due to the lack of all or a portion of such personal information data.

[0064] As used in the description above and the claims below, the phrases “at least one of A, B, or C” and “one or more of A, B, or C” include A alone, B alone, C alone, a combination of A and B, a combination of B and C, a combination of A and C, and a combination of A, B, and C. That is, the phrases “at least one of A, B, or C” and “one or more of A, B, or C” means A, B, C, or any combination thereof, such that one or more of a group of elements consisting of A, B and C, and should not be interpreted as requiring at least one of each of the listed elements A, B and C, regardless of whether A, B and C are related as categories or otherwise. Furthermore, the use of the article “a” or “the” in introducing an element should not be interpreted as being exclusive of a plurality of elements. Also, the recitation of “A, B, and/or C” is equal to “at least one of A, B, or C.” Also, the use of “a” refers to “one or more” in the present disclosure. For example, “an application” refers to “one application” or “a group of applications.”

Claims

1. A method for promoting privacy when providing query results, the method comprising, by a computing device: receiving a query that includes a unique identifier associated with a software

application, a date, and at least one parameter; generating query results based on the query, wherein the query results include one or more records; obtaining a particular random seed value based on the unique identifier and the date; generating a plurality of hash values using the unique identifier, the particular random seed value, and at least one aspect of a corresponding record of the one or more records; generating a plurality of noise values for corresponding hash values of the plurality of hash values; adjusting the one or more records using the plurality of noise values to produce one or more adjusted records; and causing information based at least in part on the one or more adjusted records to be included in a user interface that is output on a display device.

2. The method of claim 1, wherein obtaining the particular random seed value based on the unique identifier and the date includes: identifying a respective plurality of random seed values that corresponds to the date; assigning the unique identifier of the software application to a dividend value; assigning a number of random seed values in the respective plurality of random seed values to a divisor value; and performing a modulo operation on the dividend value and divisor value, wherein a remainder of the modulo operation corresponds to a position of the particular random seed value relative to other one of the respective plurality of random seed values.

3. The method of claim 1, wherein, for a given record of the one or more records, the at least one aspect includes: a metric name that corresponds to the given record; a date value that corresponds to the given record; a period value that corresponds to the given record; a grouping set value that corresponds to the given record; or some combination thereof.

4. The method of claim 1, wherein generating a given noise value of the plurality of noise values includes: generating a random value based on a given respective hash value, and a personalization value that is based on the unique identifier; and utilizing the random value to obtain the given noise value.

5. The method of claim 4, wherein: the given respective hash value is a uniformly distributed random value; the random value is a normalized random value; and the given noise value is a Gaussian distribution value.

6. The method of claim 5, wherein: the given respective hash value is generated using a SHA-256 hash function; the random value is generated using a SHA-256 hash function; and the Gaussian distribution value is based on a mean of zero and a standard deviation of two.

7. The method of claim 1, wherein adjusting the one or more records includes performing at least one operation on a given record of the one or more records based on a corresponding noise value of the plurality of noise values and a type of the given record.

8. A non-transitory computer readable storage medium configured to store instructions that, when executed by at least one processor included in a computing device, cause the computing device to promote privacy when providing query results, by carrying out steps that include: receiving at least one query that includes a unique identifier associated with a software application, a date, and at least one parameter; generating query results based on the query, wherein the query results include one or more records; obtaining a particular random seed value based on the unique identifier and the date; generating a plurality of hash values using the unique identifier, the particular random seed value, and at least one aspect of a corresponding record of the one or more records; generating a plurality of noise values for corresponding hash values of the plurality of hash values; adjusting the one or more records using the plurality of noise values to produce one or more adjusted records; and causing information based at least in part on the one or more adjusted records to be included in a user interface that is output on a display device.

9. The non-transitory computer readable storage medium of claim 8, wherein obtaining the particular random seed value based on the unique identifier and the date includes: identifying a respective plurality of random seed values that corresponds to the date; assigning the unique identifier of the software application to a dividend value; assigning a number of random seed values in the respective plurality of random seed values to a divisor value; and performing a modulo operation on the dividend value and divisor values, wherein a remainder of the modulo

operation corresponds to a position of the particular random seed value relative to other ones of the respective plurality of random seed values.

10. The non-transitory computer readable storage medium of claim 8, wherein, for a given record of the one or more records, the at least one aspect includes: a metric name that corresponds to the given record; a date value that corresponds to the given record; a period value that corresponds to the given record; a grouping set value that corresponds to the given record; or some combination thereof.

11. The non-transitory computer readable storage medium of claim 8, wherein generating a given respective noise value for a given respective hash value comprises: generating a random value based on the given respective hash value, and a personalization value that is based on the unique identifier; and utilizing the random value to obtain the given respective noise value.

12. The non-transitory computer readable storage medium of claim 11, wherein: the given respective hash value is a uniformly distributed random value; the random value is a normalized random value; and the given respective noise value is a Gaussian distribution value.

13. The non-transitory computer readable storage medium of claim 12, wherein: the given respective hash value is generated using a SHA-256 hash function; the random value is generated using a SHA-256 hash function; and the Gaussian distribution value is based on a mean of zero and a standard deviation of two.

14. The non-transitory computer readable storage medium of claim 8, wherein adjusting the one or more records includes performing at least one operation on a given record of the one or more records based on a corresponding noise value of the plurality of noise values and a type of the given record.

15. A computing device configured to promote privacy when providing query results, the computing device comprising: at least one processor; and at least one memory storing instructions that, when executed by the at least one processor, cause the computing device to carry out steps that include: receiving at least one query that includes a unique identifier associated with a software application, a date, and at least one parameter; generating query results based on the query, wherein the query results include one or more records; obtaining a particular random seed value based on the unique identifier and the date; generating a plurality of hash value using the unique identifier, the particular random seed value, and at least one aspect of a corresponding record of the one or more records; generating a plurality of noise values for corresponding hash values of s values; adjusting the one or more records using the plurality of noise values to produce one or more adjusted records; and causing information based at least in part on the one or more adjusted records to be included in a user interface that is output on a display device.

16. The computing device of claim 15, wherein obtaining the particular random seed value based on the unique identifier and the date includes: identifying a respective plurality of random seed values that corresponds to the date; assigning the unique identifier of the software application to a dividend value; assigning a number of random seed values in the respective plurality of random seed values to a divisor value; and performing a modulo operation on the dividend value and divisor values, wherein a remainder of the modulo operation corresponds to a position of the particular random seed value relative to other ones of the respective plurality of random seed values.

17. The computing device of claim 15, wherein, for a given record of the one or more records, the at least one aspect includes: a metric name that corresponds to the given record; a date value that corresponds to the given record; a period value that corresponds to the given record; a grouping set value that corresponds to the given record; or some combination thereof.

18. The computing device of claim 15, wherein generating a given respective noise value for a given respective hash value comprises: generating a random value based on the given respective hash value, and a personalization value that is based on the unique identifier; and utilizing the random value to obtain the given respective noise value.

19. The computing device of claim 18, wherein: the given respective hash value is a uniformly distributed random value; the random value is a normalized random value; and the given respective noise value is a Gaussian distribution value.

20. The computing device of claim 19, wherein: the given respective hash value is generated using a SHA-256 hash function; the random value is generated using a SHA-256 hash function; and the Gaussian distribution value is based on a mean of zero and a standard deviation of two.
